OZONE : A TOXIC SAVIOR OF LIFE

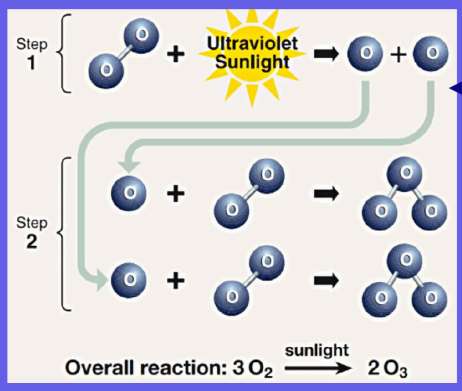
AN INSIGHT of THE ACTUAL PHENOMENON

WHAT IS OZONE ?

Ozone is a pale blue colored gas, with a distinctively pungent smell. Ozone - O_3 (trioxygen allotrope) is an inorganic molecule which is much less stable than O_2 (dioxygen allotrope) of O (Oxygen, 21% of atmosphere).
O_3 breaks down in the lower atmosphere as O_2.



NATURAL CREATION OF OZONE



Largest Ozone-production occurs in the tropical-stratosphere.

Stratospheric Ozone Production

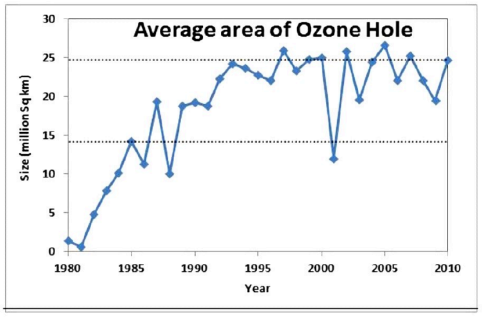
In presence of solar ultraviolet radiations (sunlight), in steps, with oxygen (O_2 , 21% of atmosphere), chemical reactions take place to form ozone molecule, O_3

GRAPH

Extreme use of CFC-gas, petrol (producing CO2) destroying ozone-layer.

Since 1979~1980 through to the early 1990s, stratospheric ozone-concentrations fell to the concerning ozone-hole level of **100 DU**; which continued approximately **around 100 DU (or below)** for several following decades...

DU(Dobson Units): Measurement unit for ozone-layer thickness in an air-column from ground to atmosphere-top



Southern hemisphere ozone hole area.

OZONE-HOLE :

Area having less than 220 DU of ozone (concentration) in the overhead column (i.e., between the ground and space)

The graph compares the size of 'Southern-Hemisphere ozone-hole' over a **20 yrs. period time, from 1980 - 2010**; where each year, **at spring, ozone hole turns largest**

INDUSTRIAL-CAUSES BEHIND OZONE-LAYER DEPLETION

A MAJOR ANTHROPOGENIC CAUSE OF OZONE-LAYER DEPLETION

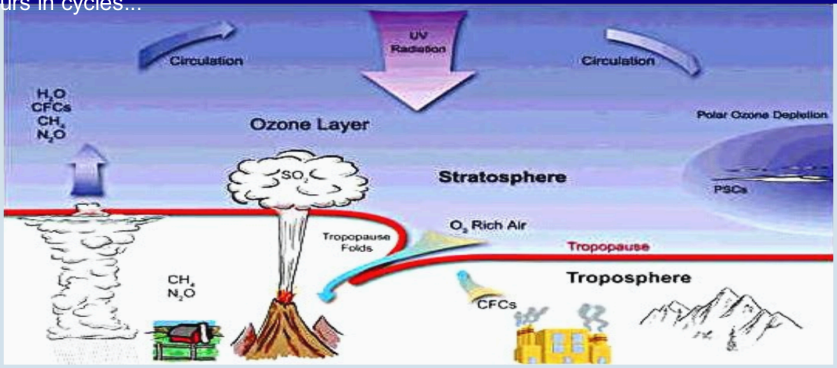
The main causes of ozone depletion and the ozone hole are manufactured synthetic chemicals, especially manufactured halocarbon refrigerants, AC-equipments, fire-extinguishers, aerosol propellants, solvents & foam-blowing agents (chlorofluorocarbons (CFCs, HCFCs, halons)), known as ozone-depleting substances (ODS).

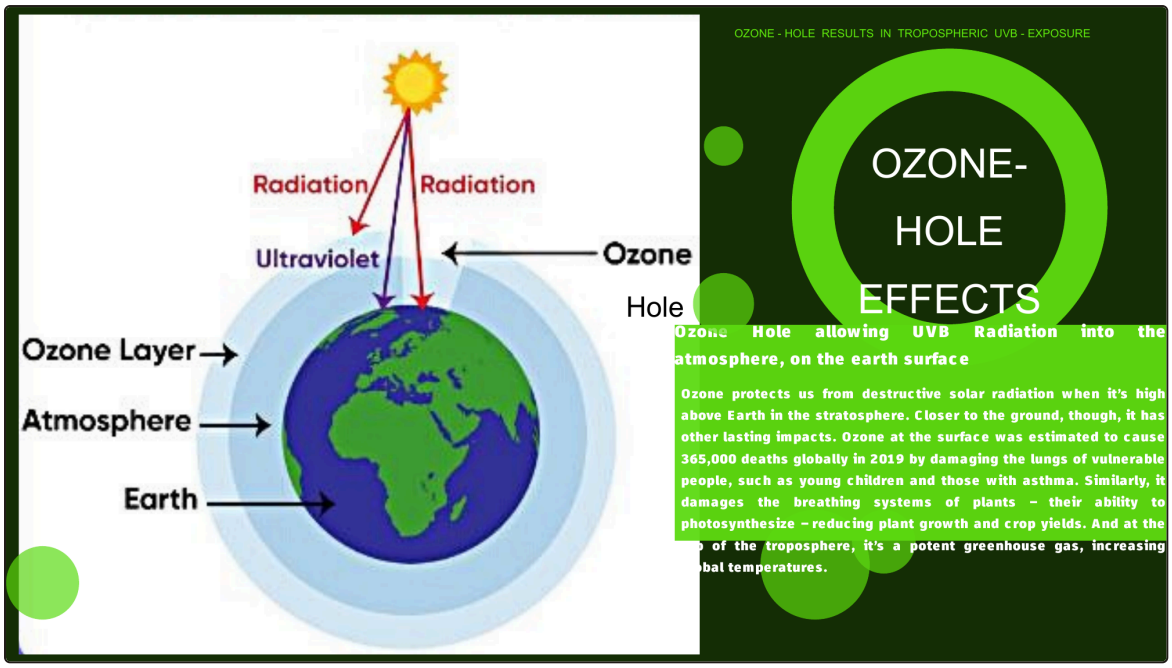
These different Industrial-Emissions produces **CFCs**, harmful **Greenhouse-Gases (CFCs & HCFCs)** that form **Chlorine & Bromine free-radicals** that are mainly responsible for breaking down O_3 to O_2, thus destroying the ozone-layer.

$O_3 \rightarrow O_2$ STRATOSPHERIC OZONE - LAYER DEPLETION

Human activities cause the emission of halogen source gases that contain chlorine and bromine atoms. These emissions into the atmosphere ultimately lead to **stratospheric ozone depletion**. **The source gases that contain only carbon, chlorine, and fluorine are called "chlorofluorocarbons," usually abbreviated as**

Once into the stratosphere, through photo-dissociation, catalytic-breakdown of ozone (O_3 to O_2) occurs in cycles...





OZONE - HOLE RESULTS IN TROPOSPHERIC UVB - EXPOSURE

OZONE-HOLE EFFECTS

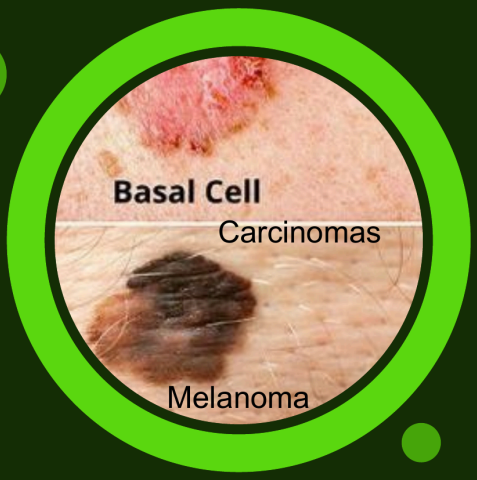
Ozone Hole allowing UVB Radiation into the atmosphere, on the earth surface

Ozone protects us from destructive solar radiation when it's high above Earth in the stratosphere. Closer to the ground, though, it has other lasting impacts. Ozone at the surface was estimated to cause 365,000 deaths globally in 2019 by damaging the lungs of vulnerable people, such as young children and those with asthma. Similarly, it damages the breathing systems of plants – their ability to photosynthesize – reducing plant growth and crop yields. And at the of the troposphere, it's a potent greenhouse gas, increasing bal temperatures.

STRATOSPHERIC OZONE DECRES & TROPOSPHERIC OZONE INCREASE

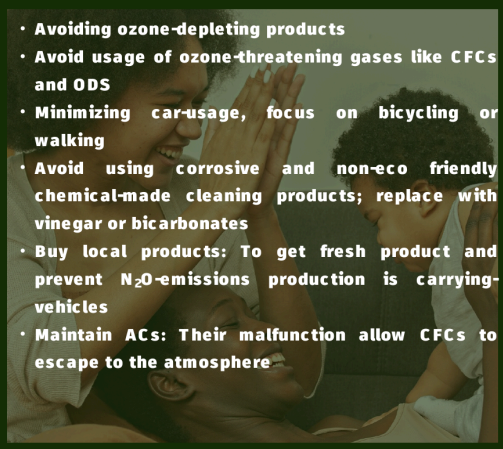
THREATS OF OZONE-IMBALANCE

- UVB-Radiation positive - increases vitamin-D production negative - sunburn, skin-cancer, cataracts
- Basal & Squamous Cell Carcinomas: causes pyridine molecules to form dimers, resulting in transcription errors while DNA-replication
- Malignant Melanoma: 19% increase (men) & 16% increase (women) in melanomas
- Cortical Cataracts
- Tropospheric-Ozone can be fatal for its highly oxidizing properties for people with asthma and breathing-difficulties
- Animals: Epidermal damages of whales in the Gulf of California associated with acute & severe sunburn have been found
- Crops: Plant growth patterns disturbed & cyanobacteria is sensitive to UVB-Radiation



BALANCED SUSTENANCE OF HUMAN & OTHER SPECIES

COMPORT COMPLYING WITH OZONE-PROTECTION



- Avoiding ozone-depleting products
- Avoid usage of ozone-threatening gases like CFCs and ODS
- Minimizing car-usage, focus on bicycling or walking
- Avoid using corrosive and non-eco friendly chemical-made cleaning products; replace with vinegar or bicarbonates
- Buy local products: To get fresh product and prevent N_2O-emissions production is carrying-vehicles
- Maintain ACs: Their malfunction allow CFCs to escape to the atmosphere




REFERENCES :
https://en.m.wikipedia.org/wiki/Ozone
https://www.nasa.gov/feature/jpl/local-lockdowns-brought-fast-global-ozone-
https://ww2.arb.ca.gov/resoucesozone-and-health
https://www.e-education.psu.edu/rocco/node/2034
https://ourworldindata.org/grapher/ozone-depleting-substance-emissions


THANK YOU